

Short summary

My research interests are efficient algorithms for mathematical problems. During my PhD, I studied the cryptanalysis of post-quantum cryptographic algorithms, in particular multivariate signature schemes. I have shown the insecurity of NIST candidates by demonstrating practical attacks that run in seconds on a laptop, and more theoretical results that improve attacks by exponential factors. All the algorithms I design are implemented and tested, with practical and reproducible results. Eager to share my knowledge, I have taught for over 230 hours during my PhD and Masters.

Experience

- 2026- Post-doctoral researcher at COSIC, KU Leuven.
Research on post-quantum cryptanalysis.
- 2022-2025 PhD Candidate at LIP6, Sorbonne Université and Thales SIX.
Publication and review of scientific papers and state-of-the-art code. Presentations at international conferences.
- 2022 Intern at Thales SIX Cryptography Lab.
- 2021 Intern at INRIA Sophia Antipolis, team COATI

Education

- 2022–2025 **Ph.D. in computer science**, Sorbonne Université, Paris
CIFRE Funding from Thales SIX and ANRT. Supervised by Simon Abelard and Mohab Safey El Din:
Geometric approach to the cryptanalysis of post-quantum signature schemes.
- 2021–2022 **Master Parisien de Recherche en Informatique**, Université Paris-Cité, Paris
- 2019–2022 **Cycle ingénieur**, Télécom Paris, Paris/Palaiseau
Specialised in applied mathematics, computer science and data science.
Received the degrees:
"Diplôme d'ingénieur de Télécom Paris" delivered by Télécom Paris.
"Diplôme de Master mention Informatique" delivered by Institut Polytechnique de Paris.

Distinctions

- 2026 Thesis Award of EDITE de Paris (1st).

Software

- 2024 Key recovery algorithm for UOV and VOX based on singular points. (SageMath)
<https://github.com/pi-r2/SingPoints>
- 2023 Polynomial-time key recovery algorithm from one vector for UOV. (SageMath)
<https://github.com/pi-r2/OneVector>
- 2021 Branch-and-bound algorithm for the computation of the pathlength of a graph. (Cython)
<https://github.com/pi-r2/pathlength>
- 2020 *with A. Barreaux* Weisfeiler-Lehman Graph Kernels in the scikit-network Python package. (Cython)
<https://github.com/sknetwork-team/scikit-network/tree/master/sknetwork/topology>

Publications in Proceedings of conferences

- 2025 **Singular points of UOV and VOX.**
Proceedings of Eurocrypt 2025, flagship international conference in cryptology.

- 2024 **One vector to rule them all: Key recovery from one vector in UOV schemes.**
Proceedings of PQCrypto 2024, the main international conference on post-quantum cryptography.
- 2023 **Shorter and Faster Identity-Based Signatures with Tight Security in the (Q)ROM from Lattices.** with E. Sageloli, C. Chevalier, P. Méaux.
Proceedings of ACNS 2023, an international conference on applied cryptography.

Student supervision

Master thesis

- 2025 **Study of a multivariate post-quantum signature scheme: $OV^\perp$.**
4 months research project for a pair of master students at Sorbonne Université.

Teaching

Algebraic Algorithms at Sorbonne Université's Master of Computer Science.

- 2023-2025 4 hour lecture on multivariate cryptology and 70 hours of exercise and lab sessions for Master students on fast computer algebra.

Computer science at Sorbonne Université's Bachelor of Computer Science.

- 2023-2025 ~105h of exercise and lab sessions in algorithmics, programming, networks, cryptology for bachelor students.

Mathematics at Lycée Janson-de-Sailly, MPSI

- 2020-2022 Designed exercise sheets and evaluated students during weekly oral interrogations in preparation of national competitive exams, for ~60 hours.

Seminars, presentations and talks

Invited talks

- August 2024 **International conference "Mathematics for post-quantum cryptanalysis"**, Budapest.
- July 2023 **Minisymposium of the international conference SIAM AAG 2023**, Eindhoven.

Invitations and seminars

- October 2025 Séminaire Canari, IMB, Bordeaux.
- October 2025 Cryptology Seminar, CWI, Amsterdam.
- June 2025 Séminaire Cryptis, XLIM, Limoges.
- May 2025 Séminaire de Cryptographie, ANSSI, Paris.
- April 2025 Effective Algebraic Geometry Seminar, IRMAR, Rennes.
- June 2024 Journée des doctorants Thales, Genevilliers.
- January 2024 ALMASTY Seminar, LIP6, Paris.

Talks at national events

- April 2025 Journées Codage et Cryptographie 2025.
- March 2025 Journées nationales du calcul formel 2025.
- March 2024 Journées nationales du calcul formel 2024.

Associations

- 2019-2021 **Technical Manager and Liaison with Crous, Aurore**
Aurore is an association providing internet access to student residences. I took part in the maintenance and deployment of our network to a new residence in a contract for CROUS Versailles.
- 2020-2022 **Administrator, Télécode**
Télécode is the programming club of Télécom Paris. (Re)created this club with fellow students P. Gimalac and J. Béguinot. In 2021, hosted an operations research contest with TotalEnergies Digital Factory.